



Divisibility of Factorials

Problem 549

The smallest number m such that 10 divides $m!$ is $m = 5$.

The smallest number m such that 25 divides $m!$ is $m = 10$.

Let $s(n)$ be the smallest number m such that n divides $m!$.

So $s(10) = 5$ and $s(25) = 10$.

Let $S(n)$ be $\sum s(i)$ for $2 \leq i \leq n$.

$S(100) = 2012$.

Find $S(10^8)$.